Classifying Chemical Reactions (p149-154)

There are many types of chemical reactions. Each type of chemical reaction has something in common. This could be common reactants or products or the way the chemical reaction takes place.

Decomposition Reactions

1. What is a decomposition reaction?

2. Write the general equation for a decomposition reaction.

3. Describe an everyday example of a decomposition reaction.

4. What is thermal decomposition and what sorts of compounds undergo thermal decomposition?

5. Describe a thermal decomposition reaction that saves lives every day.

Combination Reactions

1. What is a combination reaction?

2. Write the general equation for a combination reaction.

3. Describe a combination reaction that is important to industry.

4. Write the combination equation for the combination of hydrogen and chlorine.

Precipitation Reactions

1. Describe what happens during a precipitation reaction.

2. Give an example of a precipitation reaction.

3. What is a precipitate?

4. What does soluble mean? Give two examples of soluble substances.

5. What does insoluble mean? Give two examples of insoluble substances.

6. The solubility of ionic compounds can be predicted using some rules. Use the rules below to predict which compounds below will be soluble and which will be insoluble.
   a. Sodium chloride
   b. Silver chloride
   c. Calcium hydroxide
   d. Iron (II) nitrate
   e. Barium sulphate
   f. Magnesium iodide
   g. Sodium carbonate
   h. Calcium carbonate
   i. Copper sulphate
   j. Nickel bromide
   k. Lead bromide
   l. Sodium hydroxide

Soluble	**Exceptions**	
	Insoluble	**Slightly soluble**
Most chlorides	AgCl,	$PbCl_2$
Most bromides	AgBr,	$PbBr_2$
Most iodides	AgI, PbI_2	
All nitrates	No exceptions	
Most sulphates	$SrSO_4$, $BaSO_4$, $PbSO_4$	$CaSO_4$, Ag_2SO_4

Insoluble	**Exceptions**	
	Soluble	**Slightly soluble**
Most hydroxides	NaOH, KOH, $Ba(OH)_2$	$Ca(OH)_2$, $Sr(OH)_2$
Most carbonates	Na_2CO_3, K_2CO_3, $(NH_4)_2CO_3$	
Most phosphates	Na_3PO_4, K_3PO_4, $(NH_4)_3PO_4$	

Oxidation and Reduction

1. What is an oxidation reaction?

2. What is a combustion reaction?

3. What is a corrosion reaction?

4. What is a reduction reaction?